SECURE RAILWAY STATION NETWORK

Cisco Packet Tracer Networking & Cybersecurity Project

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Software Used: Cisco Packet Tracer

1. INTRODUCTION

This project, titled **Secure Railway Station Network**, is designed and simulated using **Cisco Packet Tracer**. It models a railway station environment in which different departments such as ticketing, administration and management require network connectivity while remaining logically separated from one another.

The project demonstrates a set of practical networking and basic cybersecurity concepts, including VLAN segmentation, Inter-VLAN routing, ICMP connectivity testing, OSPF dynamic routing, DNS name resolution, HTTP and HTTPS web services, NAT for external-side connectivity testing, SSH for secure remote administration, and restriction of insecure Telnet access.

Cisco Packet Tracer is used throughout this project to build, configure and simulate the network, and to test and verify that each configured service functions as expected. The sections that follow explain the design, configuration and testing of the network in a clear and beginner-friendly manner.

2. PURPOSE OF THE PROJECT

The purpose of this project is to design and simulate a railway station network that meets the following goals:

- Connects different railway station departments.
- Separates departments using VLANs.
- Allows controlled communication between VLANs.
- Provides internal routing between network segments.
- Provides DNS and web services for the station.
- Demonstrates secure remote administration using SSH.
- Uses NAT for external-side connectivity testing.
- Demonstrates restriction of insecure Telnet access.
- Provides a practical understanding of network security and troubleshooting.

It is important to note that this project is a **Cisco Packet Tracer simulation** created for learning purposes only, and does not represent a real railway deployment.

3. OBJECTIVES

The objectives of this project are as follows:

1. Design a railway station network using Cisco Packet Tracer.
2. Connect ticketing, administration and management devices.
3. Implement VLAN segmentation.
4. Enable communication between different VLANs through Inter-VLAN Routing.
5. Configure and verify OSPF routing.
6. Test connectivity using ICMP.
7. Configure DNS for name resolution.

8. Configure HTTP and HTTPS web services.
9. Configure NAT for external connectivity testing.
10. Configure and test secure SSH remote administration.
11. Demonstrate that Telnet remote access is restricted.
12. Understand practical network security and troubleshooting.

4. NETWORK TOPOLOGY

Figure 1 below shows the exact network topology designed and simulated in Cisco Packet Tracer for this project.

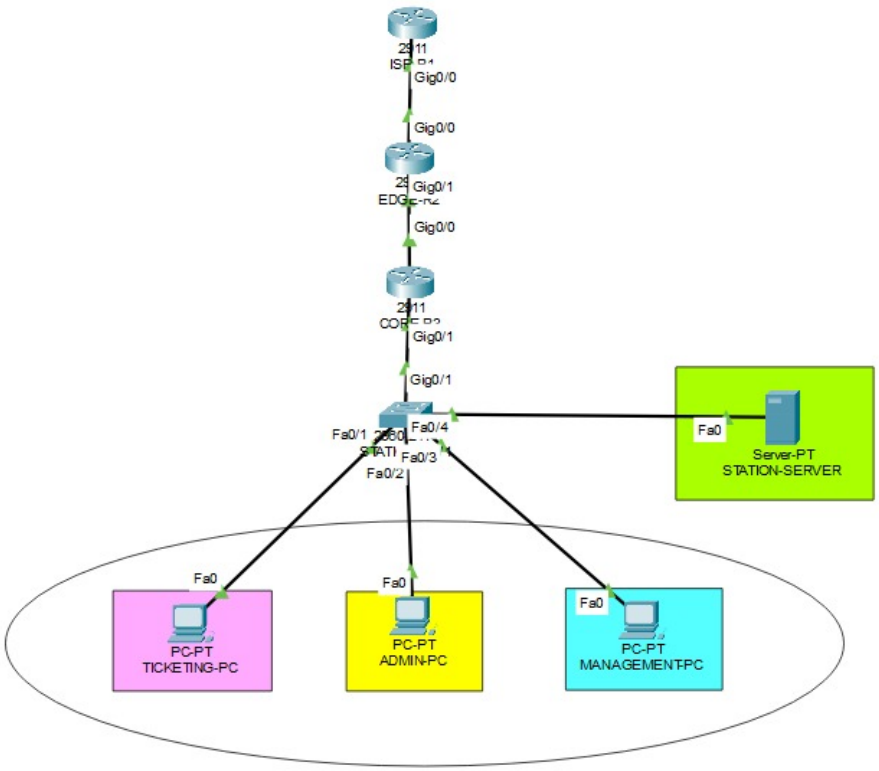


Figure 1 – Secure Railway Station Network Topology

The topology follows a layered structure: **ISP-R1** → **EDGE-R2** → **CORE-R3** → **STATION-SW1** → **Station Devices**.

ISP-R1

ISP-R1 represents the upstream/external network side of the Packet Tracer simulation.

EDGE-R2

EDGE-R2 acts as the edge router and handles NAT and external-side connectivity for the station network.

CORE-R3

CORE-R3 handles internal routing for the station network and participates in OSPF.

STATION-SW1

STATION-SW1 connects the station devices and provides VLAN-based network segmentation.

The station side of the network contains the following devices:

- Ticketing PC
- Admin PC
- Management PC
- Server

ISP-R1 represents the upstream/external network side of the Packet Tracer simulation and does not provide real Internet access.

5. DEVICES AND THEIR PURPOSE

The table below lists each device used in the topology and its role within the network.

Device	Purpose
ISP-R1	Represents the upstream/external network side
EDGE-R2	Edge router responsible for NAT and external-side connectivity
CORE-R3	Core router responsible for internal routing and OSPF
STATION-SW1	Connects station devices and provides VLAN segmentation
Ticketing PC	Represents railway ticketing operations
Admin PC	Represents railway administration
Management PC	Represents railway management
Server	Provides DNS, HTTP and HTTPS services

6. IP ADDRESSING

The IP addressing scheme used across the topology is shown below.

Device / Purpose	IP Address
Ticketing PC	192.168.30.10
Admin-side device	192.168.20.11
Server	192.168.40.10
OSPF Neighbor / EDGE-R2 address	192.168.10.1
OSPF Router ID	2.2.2.2
NAT inside-global address	10.0.0.2
External test address	8.8.8.8

These addresses are used for communication, routing, service access and connectivity testing within the Packet Tracer simulation.

7. PROTOCOLS AND TECHNOLOGIES USED

This project makes use of the following technologies and protocols.

Technology	Purpose
VLAN	Logical network segmentation
Inter-VLAN Routing	Communication between VLANs
ICMP	Connectivity testing
OSPF	Dynamic routing
DNS	Converts station.local into the server IP address
HTTP	Web communication
HTTPS	Secure web communication
NAT	Translates private/internal address for external connectivity testing
SSH	Secure remote administration
Telnet Restriction	Demonstrates restriction of insecure remote access

8. VLAN CONFIGURATION

VLANs are configured on **STATION-SW1** as shown in the table below.

VLAN	Name	Ports
VLAN 10	TICKETING	Fa0/1
VLAN 20	ADMIN	Fa0/2
VLAN 30	MANAGEMENT	Fa0/3, Fa0/4

- VLAN 10 separates ticketing traffic.
- VLAN 20 separates administration traffic.
- VLAN 30 separates management traffic.
- VLANs logically separate the different station departments from one another.

Verification

Run on: STATION-SW1

```
show vlan brief
```

The output of this command confirms the following:

- VLAN 10 TICKETING – Fa0/1
- VLAN 20 ADMIN – Fa0/2
- VLAN 30 MANAGEMENT – Fa0/3, Fa0/4

This command verifies that each VLAN has been created and that the correct ports have been assigned to it.

9. INTER-VLAN ROUTING

Because VLANs are logically separated, devices in different VLANs cannot communicate directly. Routing is required to allow this communication, and **CORE-R3** performs the internal routing function for the station network.

Demonstration

Run on: Ticketing PC → Desktop → Command Prompt

```
ping 192.168.20.11
```

Final Result: Sent = 4, Received = 4, Lost = 0 (0% loss)

- Ticketing PC is in VLAN 10.
- The destination address is on the administration side.
- Successful replies show communication between different VLANs.
- 0% packet loss indicates successful connectivity.

The first ping test may show one initial timeout due to ARP/address resolution taking place in Packet Tracer. The repeated test achieved 0% packet loss, confirming successful Inter-VLAN communication.

10. ICMP

Definition

ICMP (Internet Control Message Protocol) is used for network connectivity and diagnostic testing.

Purpose in Our Project

ICMP is used to verify communication between devices in the network.

Demonstration

The demonstration was carried out using Packet Tracer **Simulation Mode** with the following steps:

1. Switch to Simulation Mode.
2. Select/filter ICMP.
3. Use Add Simple PDU.
4. Select Ticketing PC as the source.
5. Select the destination device 192.168.20.11.

6. Use Capture/Forward.
7. Observe the packet movement.

Observed Path: Ticketing PC → Switch → CORE-R3 → destination-side switch → Admin PC

- ICMP Echo Request travels toward the destination.
- ICMP Echo Reply returns to the source.
- A successful request/reply exchange confirms connectivity.

11. OSPF

Definition

OSPF (Open Shortest Path First) is a dynamic routing protocol used to exchange routing information between routers.

Purpose in Our Project

OSPF allows the routers in the topology to learn routes to other networks dynamically.

Verification

Run on: CORE-R3

```
CORE-R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:39    192.168.10.1    GigabitEtherne
t0/0
```

The important fields in this output are explained below:

- **Neighbor ID 2.2.2.2:** OSPF Router ID of the neighboring router, EDGE-R2.
- **Pri 1:** OSPF priority.
- **FULL/BDR:** OSPF adjacency is successfully established and the neighbor is the Backup Designated Router.
- **Dead Time:** Time remaining before the neighbor is considered down if hello messages are not received.
- **Address 192.168.10.1:** Neighbor's interface address.
- **GigabitEthernet0/0:** Interface through which the neighbor is reached.

OSPF Route Verification

Run on: CORE-R3

```
CORE-R3#show ip route ospf

     8.0.0.0/32 is subnetted, 1 subnets

O       8.8.8.8 [110/3] via 192.168.10.1, 00:26:28, GigabitEthernet0/0

     10.0.0.0/30 is subnetted, 1 subnets

O       10.0.0.0 [110/2] via 192.168.10.1, 00:26:28, GigabitEthernet0/0
```

- **O** = OSPF learned route.
- **110** = OSPF administrative distance.
- **3 / 2** = OSPF metric/cost.
- **via 192.168.10.1** = next-hop address.
- **GigabitEthernet0/0** = outgoing interface.

12. DNS

Definition

DNS (Domain Name System) converts a domain/host name into an IP address.

Purpose in Our Project

DNS allows the station server to be accessed using **station.local** instead of typing the IP address directly.

Server Configuration

Configured on: **Server** → **Services** → **DNS**. DNS is **ON**.

The hostname **station.local** maps to **192.168.40.10**.

Demonstration

Run on: Ticketing PC → Desktop → Command Prompt

```
ping station.local
```

Expected Output: Pinging 192.168.40.10 with 32 bytes of data:

Final Result: Sent = 4, Received = 4, Lost = 0 (0% loss)

Result

DNS successfully resolved station.local to 192.168.40.10.

13. HTTP

Definition

HTTP (Hypertext Transfer Protocol) is used for communication between a web browser and a web server.

Purpose in Our Project

HTTP is used to access the railway station web portal hosted on the Server.

Configuration

Configured on: **Server** → **Services** → **HTTP**. HTTP is **ON**.

Demonstration

Run on: Ticketing PC → Desktop → Web Browser

```
http://station.local
```

Expected Output

The following railway station portal loads: **SECURE RAILWAY STATION – Digital Station Network Portal**.

The portal contains information related to:

- Ticketing
- Administration
- Management
- Network Architecture
- VLAN Segmentation
- OSPF Routing
- DNS
- Web Services
- SSH
- Station Network

Result

The HTTP web service was successfully accessed from the Ticketing PC.

14. HTTPS

Definition

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP and provides encrypted communication between the browser and the web server.

Purpose in Our Project

HTTPS demonstrates secure access to the railway station web portal.

Configuration

Configured on: **Server** → **Services** → **HTTPS**. HTTPS is **ON**.

Demonstration

Run on: Ticketing PC → Desktop → Web Browser

```
https://station.local
```

Expected Output

The railway station portal loads successfully.

Result

HTTPS access to the station web portal was successfully tested.

15. NAT (NETWORK ADDRESS TRANSLATION)

Definition

NAT (Network Address Translation) translates an internal/private IP address into another address for communication with an external network.

Purpose in Our Project

NAT is used to demonstrate external-side connectivity from the station network within the Packet Tracer simulation.

Demonstration

Run on: Ticketing PC → Desktop → Command Prompt

```
ping 8.8.8.8
```

Result: 4 packets received, Lost = 0%

The NAT translation was then verified on EDGE-R2.

Run on: EDGE-R2

```
EDGE-R2#show ip nat translations

Pro  Inside global     Inside local       Outside local      Outside global

icmp 10.0.0.2:15       192.168.30.10:15   8.8.8.8:15         8.8.8.8:15

icmp 10.0.0.2:16       192.168.30.10:16   8.8.8.8:16         8.8.8.8:16

icmp 10.0.0.2:17       192.168.30.10:17   8.8.8.8:17         8.8.8.8:17
```

- **Inside local:** 192.168.30.10 – private address of the Ticketing PC.
- **Inside global:** 10.0.0.2 – translated address.
- **Outside global:** 8.8.8.8 – external test address.
- **ICMP:** protocol used for the ping.
- The values :15, :16, :17 are ICMP identifiers/session entries, not TCP/UDP port numbers.

Simple Flow: 192.168.30.10 → NAT on EDGE-R2 → 10.0.0.2 → 8.8.8.8

8.8.8.8 is used here only as an external connectivity/NAT test address inside the Packet Tracer simulation, and this test does not demonstrate real Internet access.

16. SSH

Definition

SSH (Secure Shell) is a secure protocol used for remotely accessing and managing network devices.

Purpose in Our Project

SSH provides secure remote administration of the router.

Demonstration

Run on: Ticketing PC → Desktop → Command Prompt

```
ssh -l admin 192.168.20.1
```

The configured authentication details are entered when prompted.

Expected Output

The user successfully reaches the router's command-line interface/router prompt.

Result

SSH remote administration was successfully tested.

17. TELNET RESTRICTION

Definition

Telnet is a remote-access protocol that does not provide encrypted communication.

Purpose in Our Project

Telnet was tested to demonstrate that insecure remote access is restricted, while SSH is used for secure administration.

Demonstration

Run on: Ticketing PC → Desktop → Command Prompt

```
C:\>telnet 192.168.20.1

Trying 192.168.20.1 ...Open

[Connection to 192.168.20.1 closed by foreign host]
```

Expected Output

The connection reaches the router but is then closed.

Result

Telnet remote access was not permitted, while SSH was successfully used for secure remote administration.

18. SERVER SERVICES

The Server provides the following services within the network:

Service	Status
DNS	ON
HTTP	ON
HTTPS	ON

- **DNS:** Resolves station.local to 192.168.40.10.
- **HTTP:** Hosts the railway station web portal.
- **HTTPS:** Provides secure access to the railway station web portal.

19. TESTING SUMMARY

The table below summarizes all the tests carried out in this project.

Test	Run On	Result
VLAN verification	STATION-SW1	VLANs correctly displayed
Inter-VLAN Ping	Ticketing PC	4/4 successful, 0% loss
ICMP Simulation	Packet Tracer Simulation Mode	Echo Request and Reply observed
OSPF Neighbor	CORE-R3	FULL/BDR adjacency
OSPF Routes	CORE-R3	OSPF routes displayed
DNS	Ticketing PC	station.local resolved to 192.168.40.10
HTTP	Ticketing PC browser	Station portal loaded
HTTPS	Ticketing PC browser	Station portal loaded
NAT	Ticketing PC + EDGE-R2	Translation verified
SSH	Ticketing PC	Secure router access successful
Telnet	Ticketing PC	Connection closed/restricted

20. PRACTICAL CHALLENGES AND SOLUTIONS

Several practical challenges were encountered during the project, along with the solutions applied.

VLAN Communication

Challenge: Different VLANs cannot communicate directly.

Solution: Inter-VLAN routing was configured so communication between required VLANs could be tested.

Initial Ping Timeout

Challenge: The first connectivity test may show a timeout.

Solution: Repeat the test after initial address/ARP resolution. The final test achieved 0% packet loss.

Understanding OSPF

Challenge: OSPF output contains fields such as Neighbor ID, State, Dead Time and metric.

Solution: Verify the OSPF neighbor and routing table using the appropriate show commands.

DNS Name Resolution

Challenge: Using a hostname requires correct DNS configuration.

Solution: Configure the server's DNS service and map station.local to 192.168.40.10.

Secure Remote Access

Challenge: Telnet is not secure.

Solution: SSH was tested for secure remote administration, while Telnet access was restricted.

21. INDUSTRY RELEVANCE – RAILWAY STATIONS

The concepts demonstrated in this project relate to real railway station network environments in the following ways:

Ticketing Systems

Ticketing systems require reliable network connectivity for station operations.

Department Segmentation

Ticketing, administration and management can be logically separated using VLANs.

Inter-Department Communication

Inter-VLAN routing can allow required communication between different railway station departments.

Station Web Services

DNS, HTTP and HTTPS can support internal web-based station services and information portals.

Network Routing

OSPF demonstrates dynamic routing concepts that can be relevant when multiple network segments or routers need to exchange routing information.

Secure Administration

SSH demonstrates secure remote administration of network devices.

Insecure Protocol Restriction

The Telnet test demonstrates why insecure remote-access methods should be restricted in environments where network security is important.

External Connectivity

NAT demonstrates how internal station network addresses can be translated for external-side communication.

Reliability and Troubleshooting

ICMP and routing verification provide basic methods for checking connectivity and troubleshooting network problems.

This project is a Cisco Packet Tracer simulation created for educational purposes and is not a real railway network deployment.

22. SECURITY FEATURES

This project demonstrates the following security-related concepts:

VLAN Segmentation

Separates ticketing, administration and management traffic logically.

HTTPS

Provides encrypted communication for the web portal.

SSH

Provides secure remote administration.

Telnet Restriction

Demonstrates restricting insecure remote administration.

Network Segmentation

Separating departments reduces unnecessary direct communication between network segments.

Monitoring and Testing

ICMP, OSPF verification and NAT translation checks help identify connectivity and routing issues.

23. CONCLUSION

This project successfully demonstrated a secure railway station network simulation using Cisco Packet Tracer. It covered VLAN segmentation, Inter-VLAN routing, ICMP connectivity testing, OSPF routing, DNS, HTTP, HTTPS, NAT, SSH, and Telnet restriction.

The tests carried out confirmed connectivity, routing, name resolution, web services, NAT translation and secure remote administration within the Packet Tracer simulation. This project therefore provided practical, hands-on experience in fundamental networking and cybersecurity concepts, without representing a real-world deployment.